Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 24: Split the Problem

Proof by Cases

Chartrand & Zhang, Section 3.4

Name: _____ Date: _____

Where we are

Direct proof works when the hypothesis hands you an equation. Contrapositive works when the *other* hypothesis does. Today: what to do when neither does — because the hypothesis does not pin the situation down enough to compute with.

By the end of today you should be able to

1. Recognise when a proof needs cases.
2. Choose cases that are exhaustive.
3. Use "without loss of generality" correctly.

1. The idea

Proof by cases

To prove $\forall x \in S,\ R(x)$, split S into two or more subsets covering all of S — typically a **partition** — and prove $R(x)$ separately in each.
The extra information about *which* subset x lies in is what makes each case tractable.

The cases must cover everything

If your cases miss even one element of S, you have proved nothing about that element and the proof is incomplete.
For integers, the standard split is **even or odd** — and it works because every integer is exactly one of the two. That is a partition, from Lecture 14.

2. Worked: Result 3.26

Result 3.26

If n is an integer, then $n^2 - n$ is even.

Why not direct? "n is an integer" gives you no equation to substitute. The hypothesis is too weak on its own.

Proof. Let n be an integer. We consider two cases.

Case 1: n is even. Then $n = 2a$ for some integer a, and

$$n^2 - n = 4a^2 - 2a = 2\big(2a^2 - a\big).$$

Since $2a^2 - a$ is an integer, $n^2 - n$ is even.

Case 2: n is odd. Then $n = 2b + 1$ for some integer b, and

$$n^2 - n = (4b^2 + 4b + 1) - (2b + 1) = 4b^2 + 2b = 2\big(2b^2 + b\big).$$

Since $2b^2 + b$ is an integer, $n^2 - n$ is even.

Since every integer is even or odd, the result holds for all n. ■

A shorter route, for comparison

$n^2 - n = n(n-1)$, a product of two consecutive integers, so one of them is even and the product is even.
Both proofs are correct. **Noticing a factorisation often beats splitting into cases** — but cases always work, and that reliability is why the technique is worth having.

3. Parity of two integers

Same and opposite parity

Integers m and n are of the **same parity** if both are even or both are odd; otherwise they are of **opposite parity**.

Every two integers are of the same parity or of opposite parity, never both — so the negation of "same parity" is "opposite parity".

Result 3.27. *Let $m, n \in \mathbb{Z}$. Then $3m + 5n$ is even if and only if m and n are of the same parity.*

This needs two implications, and each one needs two cases — four arguments in all.

Direction	Cases
same parity $\Rightarrow$ $3m + 5n$ even	both even; both odd
$3m + 5n$ even $\Rightarrow$ same parity	by contrapositive: m even & n odd; m odd & n even

Independent letters in independent cases

In the case "both even" write $m = 2a$ and $n = 2b$. Writing $m = 2a$ and $n = 2a$ would assert $m = n$, which the claim never said.
Reusing a letter across *different* cases is fine — the cases are independent. Reusing it for two different quantities *within* a case is not.

4. Without loss of generality

Sometimes two cases have identical arguments with the names swapped. Then you may prove one and say **"without loss of generality."**

Result 3.28. *Let $m, n \in \mathbb{Z}$. Then mn is odd if and only if m and n are both odd.*

For the reverse direction we prove the contrapositive: if m and n are not both odd, then mn is even. By De Morgan, "not both odd" means at least one is even.

Assume without loss of generality that m is even, so $m = 2a$. Then $mn = 2a\,n = 2(an)$, which is even.

When you may not say it

"Without loss of generality" is legitimate only when the cases are **genuinely symmetric** — when swapping the names turns one argument into the other, word for word.
If the two cases differ in any real way, you must write both. Using the phrase to skip an argument you have not checked is a common and costly habit.

5. Cases beyond parity

Other natural splits:

Situation	Typical cases
A claim about $\lvert x \rvert$	$x \geq 0$ and $x < 0$
A claim about $\mathbb{Z}$ and division by 3	remainder 0, 1, or 2
A hypothesis that is a disjunction	one case per disjunct
Sets: $x \notin B \cap C$	$x \notin B$ or $x \notin C$

That last row is the one you have already used — in Lecture 14 and again on the midterm. A disjunctive hypothesis *is* an invitation to split.

Exit ticket

1. Why must the cases in a proof by cases cover the whole domain?

2. Prove by cases: for every integer n, $n^2 + n$ is even.

Before Monday

Read Section 3.5, "Counterexamples," pp. 102–105.

Self-check (not collected). Section 3.4, Exercises 1, 3, 5, and 9 — odd-numbered, answers in the back.

Coming up

Homework 9 is due Friday, October 23 — Sections 5.5, 3.2, and 3.3.
Midterms will be returned as soon as they are graded.

Questions I still have